

A PROJECT REPORT

ON

VisionVox

Submitted in partial fulfillment of the requirements
of the degree of

BACHELOR OF ENGINEERING

Computer Science & Engineering

(Artificial Intelligence & Machine Learning)

by

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CERTIFICATE

This is to certify that the Mini Project entitled “ **Vision Vox** ” is a bonafide work of Asmita Pokharkar, Aditya Chaferkar, Harsh Khanvilkar and Ankit Pal submitted in partial fulfillment of the requirement for the award of the degree of “**Bachelor of Engineering**” in “**Computer Science & Engineering (Artificial Intelligence & Machine Learning)**”.

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MINI PROJECT APPROVAL

This Mini Project entitled “Vision Vox ” by Asmita Pokharkar, Aditya Chaferkar, Harsh Khanvilkar and Ankit Pal is approved for the degree of **Bachelor of Engineering** in “**Computer Science & Engineering (Artificial Intelligence & Machine Learning)**”

Examiner:

1.....

2.....
(External Examiner name & Sign)

Date:

Place:

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principals of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/ data / fact / source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

VisionVox is an intelligent assistive system designed to enhance accessibility for visually impaired and speech-impaired individuals by leveraging advancements in artificial intelligence and computer vision. The system integrates real-time object detection, gesture recognition, and text-to-speech technologies into a unified platform that aids both navigation and communication.

The navigation module utilizes a camera-based input and deep learning models to detect surrounding objects and obstacles, providing immediate auditory feedback to the user. Simultaneously, the gesture recognition module interprets predefined hand gestures and converts them into meaningful speech outputs, enabling effective communication for users with speech impairments. Developed using technologies such as Python, OpenCV, and TensorFlow, the system ensures real-time performance with reasonable accuracy while maintaining a cost-effective and scalable design. VisionVox aims to bridge the gap between assistive needs and technological solutions by offering a user-friendly, portable, and efficient system.

This project demonstrates the practical application of AI in assistive technologies and highlights its potential to significantly improve independence and quality of life for differently-abled individuals.

Acknowledgment

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TABLE OF CONTENTS

Abstract.....	I
Acknowledgment.....	II
Table of contents.....	III

Chapter 1. Introduction

1.1 Introduction.....	
1.2 Motivation.....	
1.3 Statement of the Problem	

Chapter 2. Literature Survey

2.1 Survey Existing System (referenced research paper)	
2.2 Limitations of Existing system or research gap.....	
2.3 Objective	
2.4 Scope of the Work	

Chapter 3. Proposed System

3.1 Details of Hardware & Software.....	
3.2 Design details.....	
3.3 Methodology	

Chapter 4. Results Analysis

Chapter 5. Conclusion & Future scope

Chapter 6. References

Chapter 1. Introduction

1.1 Introduction

Assistive technologies have become increasingly important in improving the quality of life for individuals with physical disabilities. With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), it is now possible to develop intelligent systems that can perceive, interpret, and respond to real-world environments. VisionVox is an AI-based assistive system designed to support visually impaired and speech-impaired individuals by combining navigation assistance and communication capabilities into a single platform.

The system utilizes computer vision techniques to analyze real-time video input and detect objects in the surroundings, thereby helping users navigate safely. In addition, it incorporates a gesture recognition module that interprets hand gestures and converts them into speech using text-to-speech technology. This dual functionality makes VisionVox a comprehensive assistive solution aimed at enhancing independence and interaction.

1.2 Motivation

The primary motivation behind developing VisionVox is to address the everyday challenges faced by visually and speech-impaired individuals. Many existing assistive devices are either expensive, limited in functionality, or not easily accessible to the general population.

Key motivating factors include:

- The need for affordable and accessible assistive solutions
- Enhancing independence and confidence among differently-abled individuals
- Leveraging AI to solve real-world accessibility problems
- Integrating multiple assistive features into a single system
- Providing real-time support for navigation and communication

This project aims to create a practical and impactful solution that can be widely adopted.

1.3 Statement of the Problem

Visually impaired individuals face significant challenges in navigating their surroundings due to the inability to perceive obstacles and environmental changes. Similarly, speech-impaired individuals often struggle to communicate effectively with others, leading to social and practical limitations.

Existing solutions typically focus on a single problem domain and lack integration, making them less efficient and more cumbersome to use. Additionally, many systems are costly and not scalable for widespread use.

Therefore, there is a need to develop an integrated, cost-effective, and intelligent system that can assist users in both navigation and communication using real-time AI-based technologies.

VisionVox aims to address this gap by providing a unified solution that enhances accessibility, usability, and independence.

Chapter 2. Literature Survey

2.1 Survey of Existing Systems (Referenced Research Work)

Several research works have been carried out in the domain of assistive technologies using Artificial Intelligence, particularly focusing on object detection and gesture recognition.

A study on deep learning-based assistive systems highlights that modern object detection models significantly improve navigation for visually impaired individuals by identifying obstacles and providing auditory feedback. These systems use deep learning algorithms to detect and classify objects in real-time, enhancing user safety and independence .

Recent research (2025) proposes a smartphone-based assistive system using TensorFlow and SSD MobileNet, capable of detecting multiple objects and converting them into speech output. This system demonstrates the effectiveness of integrating real-time object detection with text-to-speech for navigation assistance . In the field of gesture recognition, studies show that gesture-based interaction is a natural and intuitive method of communication in human-computer interaction systems. Gesture recognition techniques, including visual sensing and machine learning approaches, have gained popularity due to their flexibility and ease of use .

Additionally, research on intelligent gesture recognition systems indicates that deep learning models such as convolutional neural networks (CNNs) are widely used to classify gestures with high accuracy. These systems are especially useful in applications such as sign language interpretation and assistive communication .

Recent integrated systems also combine gesture control with object detection, demonstrating the potential of multi-modal assistive technologies. For example, AI-based smart assistive devices use gesture input along with object detection and voice feedback to enhance user interaction and autonomy.

2.2 Limitations of Existing Systems / Research Gap

Despite significant advancements, existing systems have several limitations:

- Most systems focus on either navigation or communication, not both
- High computational requirements make real-time implementation difficult
- Accuracy is affected by environmental factors such as lighting and background noise
- Gesture recognition systems often lack robustness in dynamic environments
- Many solutions are expensive and not easily accessible

- Lack of integration between multiple assistive technologies

Research also highlights that gesture recognition systems still face challenges related to stability, adaptability, and real-world performance, especially under varying conditions .

Thus, there exists a gap for a cost-effective, integrated, and real-time assistive system.

2.3 Objectives

The main objectives of the proposed system VisionVox are:

- To develop a real-time object detection system for navigation assistance
- To implement a gesture recognition module for communication
- To convert detected outputs into audio feedback using text-to-speech
- To create an integrated platform combining navigation and communication
- To design a cost-effective and user-friendly solution
- To ensure real-time performance with acceptable accuracy

2.4 Scope of the Work

The scope of this project includes the design and development of an AI-based assistive system with the following capabilities:

- Real-time object detection using camera input
- Gesture recognition for communication support
- Audio feedback generation using text-to-speech
- Deployment on devices such as laptops or smartphones

The system is primarily intended for:

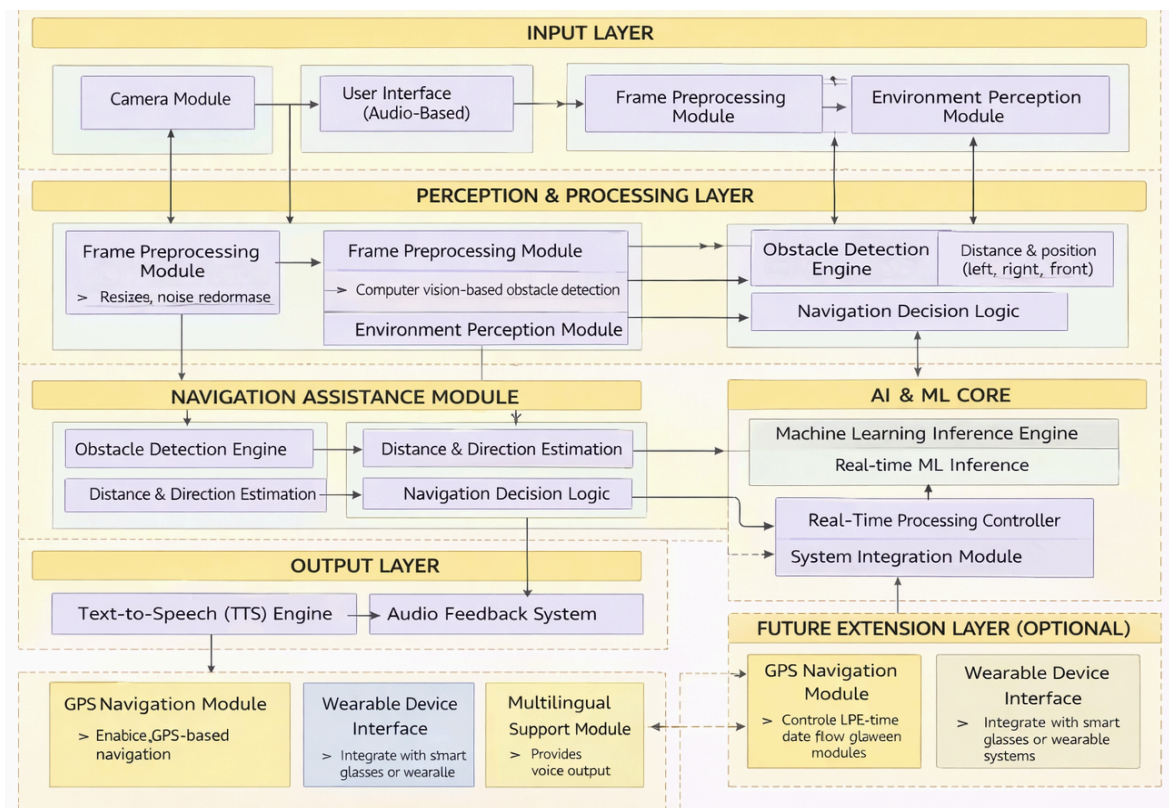
- Visually impaired individuals for navigation assistance
- Speech-impaired individuals for communication

Future enhancements may include:

- Mobile application deployment
- GPS-based outdoor navigation
- Multi-language voice support
- Integration with wearable devices

Chapter 3. Proposed System

- **Camera-Based Perception Module:** Captures real-time video of surroundings and hand gestures to serve as input for obstacle detection and gesture recognition.
- **Obstacle Detection Module:** Identifies nearby obstacles using computer vision techniques to assist safe navigation.
- **Distance & Direction Analysis:** Determines relative position of obstacles (left, right, front) to generate meaningful navigation guidance.
- **Audio Navigation Guidance:** Provides real-time voice alerts and instructions for visually impaired users.
- **Gesture-to-Text Mapping:** Converts recognized hand gestures into predefined text messages.
- **Text-to-Speech Conversion Module:** Transforms text output into clear and understandable voice messages.

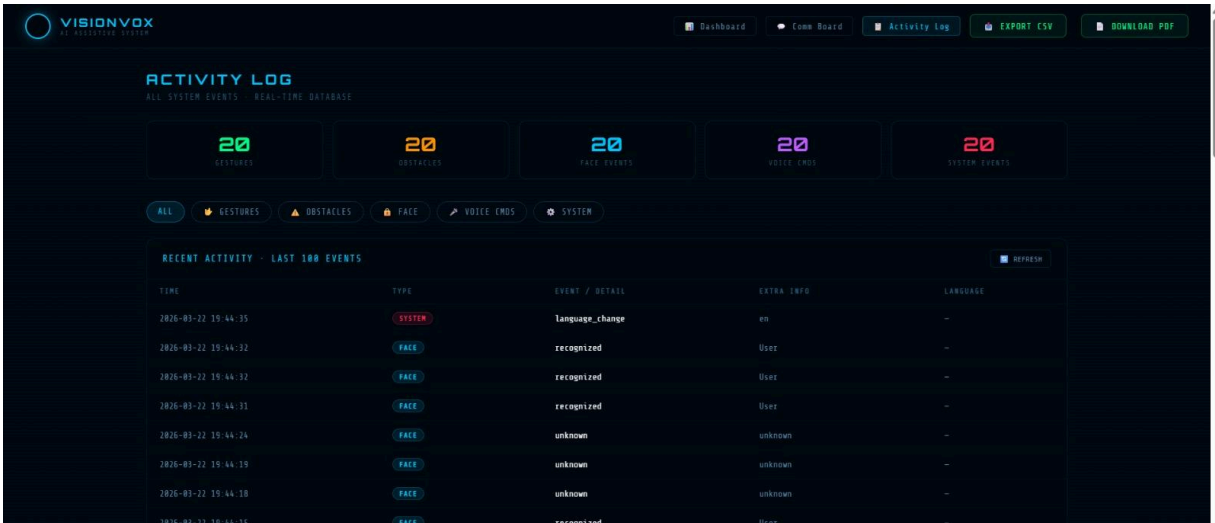
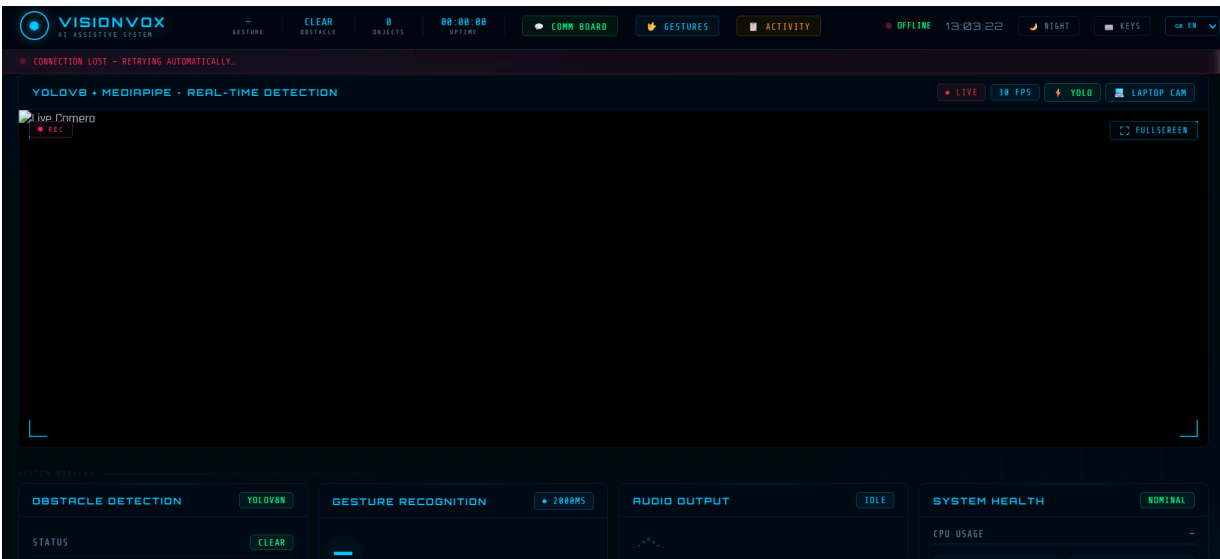


3.1 Proposed system

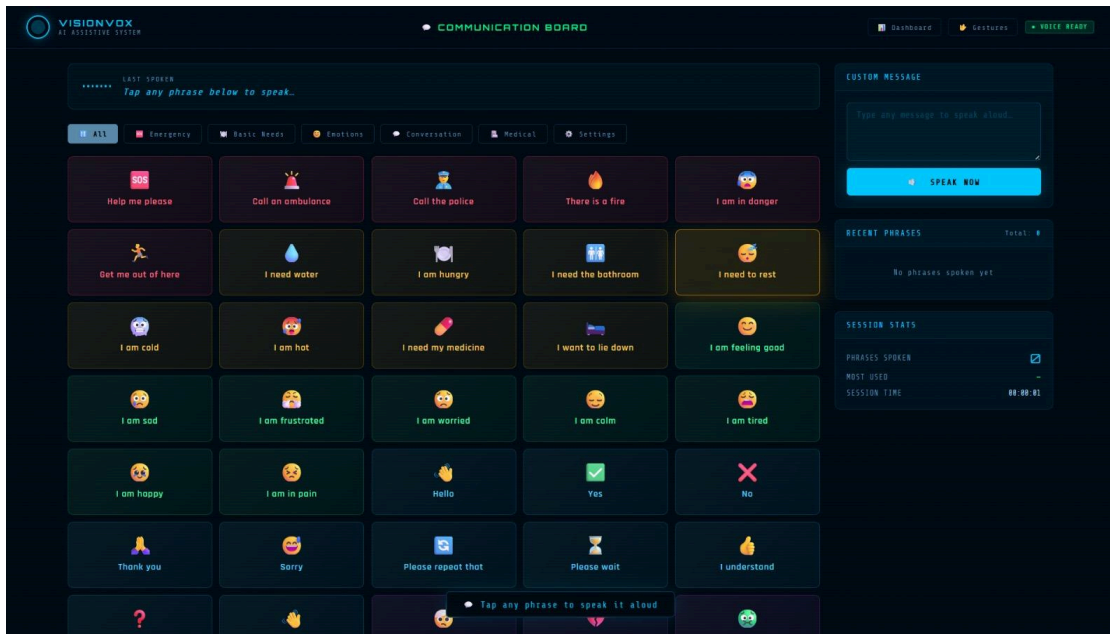
Chapter 4. Results Analysis



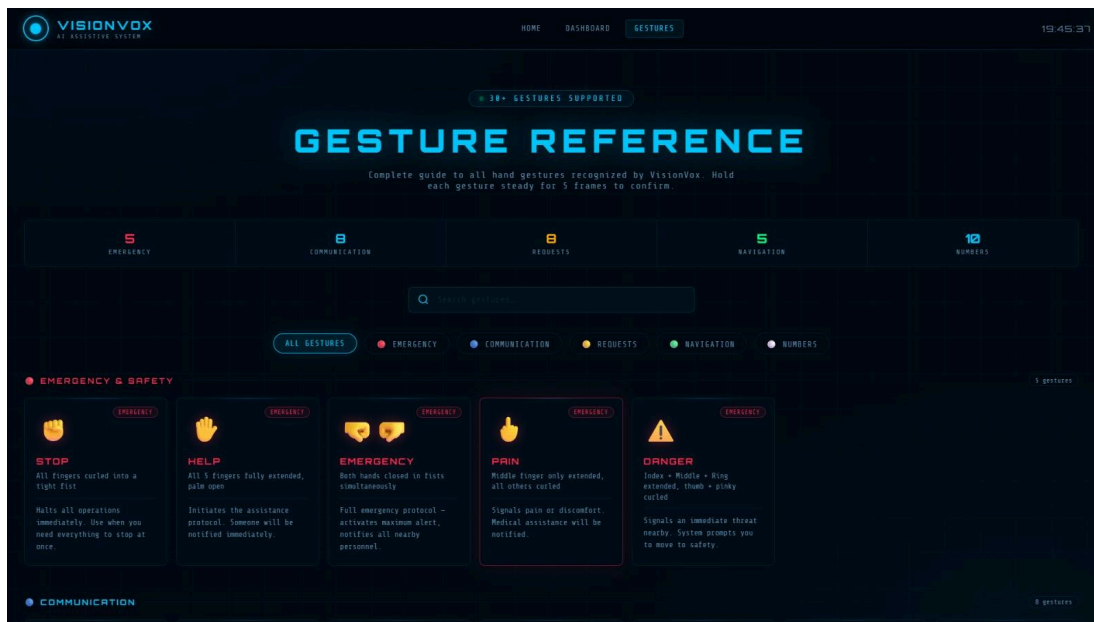
4.1 landing page



4.2 and 4.3 camera dashboard



4.4 communication dashboard



4.5 Gesture Board

Chapter 5. Conclusion & Future scope

5.1 Conclusion

The VisionVox system successfully demonstrates the application of Artificial Intelligence in developing an integrated assistive solution for visually impaired and speech-impaired individuals. By combining real-time object detection and gesture recognition with text-to-speech technology, the system provides both navigation assistance and communication support within a single platform. The project achieves its primary objectives of creating a cost-effective, user-friendly, and efficient system capable of operating in real-time conditions. The implementation using technologies such as OpenCV, TensorFlow, and MediaPipe ensures reliable performance with satisfactory accuracy. The modular architecture allows flexibility and scalability, making it suitable for further enhancements.

Overall, VisionVox highlights the potential of AI-driven assistive technologies in improving independence, accessibility, and quality of life for differently-abled individuals.

5.2 Future Scope

Although the current system performs effectively, there are several opportunities for improvement and expansion:

- **Mobile Application Deployment:**
Convert the system into an Android/iOS application for better portability and real-world usability
- **GPS-Based Navigation:**
Integrate location tracking for outdoor navigation and route guidance
- **Improved Accuracy:**
Use advanced deep learning models to enhance object detection and gesture recognition accuracy
- **Wearable Device Integration:**
Deploy the system on smart glasses or wearable devices for hands-free operation
- **Cloud Integration:**
Use cloud computing for faster processing and data storage

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